Recitation 9

Probability

Expected Value

Intuitively, the expected value is the weighted average of values, kind of a mass center of the probability distribution.

More formally, the *expected value* of a random variable is denoted $\mathbb{E}[X]$ and is defined as

$$\mathbb{E}[X] = \sum_{s \in S} X(s) \Pr(s) = \sum_{r \in X(S)} r \Pr(X = r).$$

We define the *conditional expected values* as follows: Given that event E has occurred, the expectation of random variable X is

$$\mathbb{E}[X \mid E] = \sum_{r \in X(S)} r \Pr(X = r \mid E). \tag{1}$$

Moreover, the *linearity of expectation* can be very useful in calculating expected value: Given that Z, X, Y are three random variables defined on a sample space S and a and b are two real numbers such that $Z = aX + bY$, we know that $\mathbb{E}[Z] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ must be true.

Let's practice this through a task:

Task 1

Tim and Joe are playing a game. They flip a fair coin 3 times. When the coin is heads, Joe pays \$1 to Tim; and when the coin is tails, Tim pays \$1 to Joe.

a. Let G_i be a random variable representing what Tim gains on the i-th round. For instance, $G_3 = -1$ if the coin is tails.

   What is the expected value of G_i?

Since $\Pr(G_i = 1) = \frac{1}{2}$ and $\Pr(G_i = -1) = \frac{1}{2}$, $\mathbb{E}[G_i] = \frac{1}{2} - \frac{1}{2} = 0$.

b. Let G be a random variable that represents Tim's *total* gain in this game. What is the expected value of G?

$\mathbb{E}[G] = \mathbb{E}[G_1] + \mathbb{E}[G_2] + \mathbb{E}[G_3]$, where G_i is the expected gain from flip i. We already know that $\mathbb{E}[G_i] = 0$, so $\mathbb{E}[G] = 0 + 0 + 0 = 0$ as well.

c. What is the expected value of G if the coin is biased and the probability of heads is p? in other words, generalize your solution from part b in terms of p.

Now, $\mathbb{E}[G_i] = p \cdot 1 + (1 - p) \cdot (-1) = p - 1 + p = 2p - 1$. Thus, $\mathbb{E}[G] = \mathbb{E}[G_1] + \mathbb{E}[G_2] + \mathbb{E}[G_3] = (2p - 1) + (2p - 1) + (2p - 1) = 6p - 3$.

d. Tim and Joe are still using the biased coin from part c. Let H_1 be the event that the first coin is heads. What is $\mathbb{E}[G|H_1]$?

$\mathbb{E}[G \mid H_1] = 1 + (2p - 1) + (2p - 1) = 4p - 1$.

G_1 is 1 from the first round guaranteed, and the other terms are from part c.

e. Use your answers to calculate $\mathbb{E}[G]$ and $\mathbb{E}[G \mid H_1]$ when $p = 0.7$.

$\mathbb{E}[G] = 6p - 3 = 6 \cdot 0.7 - 3 = 1.2$

$\mathbb{E}[G \mid H_1] = 4p - 1 = 4 \cdot 0.7 - 1 = 1.8$

f. Assume that $p = 0.7$ and let's say we want to change the game to make it "fair." If the flip is tails, then Tim pays a dollar to Joe—how much should Joe pay Tim on Heads so that for any number of flips we know $\mathbb{E}[G] = 0$?

Each $\mathbb{E}[G_i]$ is $0.3 \cdot (-1) + 0.7 \cdot x$, where x is the number of dollars that Joe should pay Tim on Heads. Thus, we have $\mathbb{E}[G] = (0.3 \cdot (-1) + 0.7 \cdot x) \cdot 3 = 0$. Solving for x, we have $x = \frac{3}{7}$. So, Joe should pay Tim \$0.43 on heads so that the game is fair.

Task 2

Tim and Joe are now playing a similar, but different, game. This time they flip a coin **2 times**. Let X be the random variable that is equal to the number of heads and Y the random variable that is equal to the number of tails. At the end of the game, Joe pays Tim X^2 dollars. Once again, let G be the random variable for Tim's gain.

a. Calculate $\mathbb{E}[X]^2$ when the probability of heads is $p = 0.7$.

$\mathbb{E}[X] = 0.7 + 0.7 = 1.4$. So, $\mathbb{E}[X]^2 = 1.4^2 = 1.96$

b. Calculate $\mathbb{E}[G]$ when $p = 0.7$.

$\mathbb{E}[G] = \mathbb{E}[X^2] = (0.7)^2(4) + \binom{2}{1}(0.3)(0.7)(1) = 2.38$

c. Does $\mathbb{E}[G] = \mathbb{E}[X]^2$?

No!

In general for any random variable with nonzero variance we have $\mathbb{E}[X^2] > \mathbb{E}[X])^2$. But you will learn about variance in more advanced courses!

d. Find an example that shows that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ does not hold where X and Y are not independent variables.

Hint: Try X and Y as defined above.

We know that X and Y, as defined in this problem, are not independent variables. We know that $\mathbb{E}[X]\mathbb{E}[Y] = 1 \cdot 1 = 1$. We can also calculate $\mathbb{E}[XY]$ to be $(\frac{1}{4})(2)(0) + (\frac{1}{2})(1)(1) = (\frac{1}{4})(0)(2) = \frac{1}{2}$.

Thus, this is an example of where X and Y are dependent variables where $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$.

e. Prove that if X and Y are two independent random variables, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Assume X and Y are independent. So for any x and y in their range we have $\Pr(X = x \wedge Y = y) = \Pr(X = x)\Pr(Y = y)$. We now use the definition of expected value.

$$\begin{aligned}
\mathbb{E}[XY] &= \sum_{x \in X(S)} \sum_{y \in Y(S)} xy \Pr(X = x \wedge Y = y) && (2) \\
&= \sum_{x \in X(S)} \sum_{y \in Y(S)} xy \Pr(X = x)\Pr(Y = y) && (3) \\
&= \left(\sum_{x \in X(S)} x \Pr(X = x) \right) \left(\sum_{y \in Y(S)} y \Pr(Y = y) \right) && (4) \\
&= \mathbb{E}[X]\mathbb{E}[Y] && (5)
\end{aligned}$$

Bayes Rule

The Bayes Rule can be summarized as

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where A and B are events and $P(B) \neq 0$.

Task 3

Assume Brown's CS department has an evaluation system for CS courses based on student evaluations. In any class, the students can fill the evaluation form and give a score of $0, 1$, or 2 to the course. Let X be the random variable of this score. The students of CS0220 either like the course with probability $3/4$ (Event L) or they do not like the course with probability $1/4$ (Event $\neg L$).

Assume that the conditional probability distribution of X given L is

$$\Pr(X = 0 \mid L) = 1/8$$

$$\Pr(X = 1 \mid L) = 1/4$$
$$\Pr(X = 2 \mid L) = 5/8$$

and given that they do not like the course ($\neg L$) it is

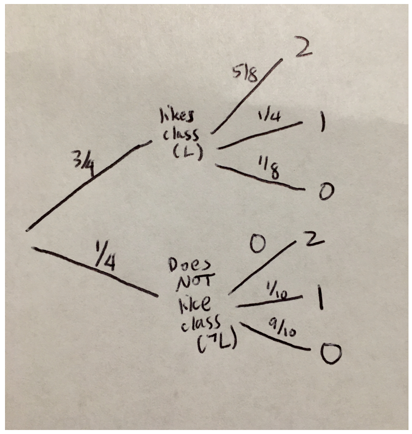
$$\Pr(X = 0 \mid \neg L) = 9/10$$

$$\Pr(X = 1 \mid \neg L) = 1/10$$
$$\Pr(X = 2 \mid \neg L) = 0.$$

a. If a student has given score of 0 to CS0220, what is the probability that they do not like the course?

$\Pr(\neg L \mid X = 0) = \frac{\Pr(X=0|\neg L)\Pr(\neg L)}{\Pr(X=0)} = \frac{\frac{9}{10}\cdot\frac{1}{4}}{\frac{1}{4}\cdot\frac{9}{10}+\frac{3}{4}\cdot\frac{1}{8}} = \frac{12}{17}$. The formula for conditional probability $\Pr(\neg L \mid X = 0) = \frac{\Pr(\neg L \cap X=0)}{\Pr(X=0)}$ gives the same answer.

Note: Drawing out a diagram like this may be helpful for this problem:



b. Use the definition of conditional expected value (Equation 1) and find $\mathbb{E}[X \mid \neg L]$

$\mathbb{E}[X \mid \neg L] = \sum r \Pr(X = r \mid \neg L) = 2 \cdot 0 + 1 \cdot \frac{1}{10} + 0 \cdot \frac{9}{10} = \frac{1}{10}$.

c. *Optional:* Find $\mathbb{E}[X]$.

$$\begin{aligned}\mathbb{E}[X] =& 2\Pr(X = 2) + 1\Pr(X = 1) + 0\Pr(X = 0)\\ =& 2\left(\Pr(X = 2 \mid L)\Pr(L) + \Pr(X = 2 \mid \neg L)\Pr(\neg L)\right)\\ &+ \Pr(X = 1 \mid L)\Pr(L) + (\Pr(X = 1 \mid \neg L)\Pr(\neg L)\\ &= 2 \cdot \frac{3}{4} \cdot \frac{5}{8} + 1 \cdot \frac{3}{4} \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} \cdot \frac{1}{10} = \frac{23}{20} = 1.15.\end{aligned}$$

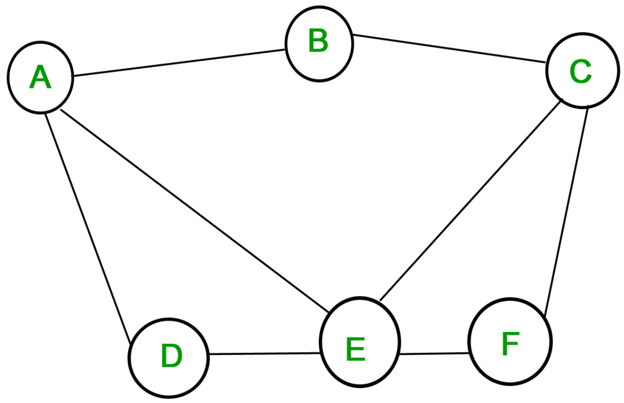
Alternative solution: $\mathbb{E}[X] = \Pr(L)\mathbb{E}[X \mid L] + \Pr(\neg L)\mathbb{E}[X \mid \neg L] = \frac{3}{4} \cdot (\frac{5}{8} \cdot 2 + \frac{1}{4} \cdot 1 + \frac{1}{8} \cdot 0) + \frac{1}{4} \cdot (0 \cdot 2 + \frac{1}{10} \cdot 1 + \frac{9}{10} \cdot 0) = \frac{23}{20} = 1.15$

Checkpoint — Call over a TA!

Introduction to Graph Theory

A **graph** is two related sets: $V(G)$, the vertex set, and $E(G)$, the edge set. Each element of $E(G)$ is a set containing exactly two elements of $V(G)$.

We often visualize graphs by drawing the vertices as dots and the edges as lines between them. Here is an example of a graph with vertex set $\{A, B, C, D, E, F\}$ and edge set $\{\{A, B\}, \{B, C\}, \{A, E\}, \{C, E\}, \{A, D\}, \{C, F\}, \{D, E\}, \{E, F\}\}$.



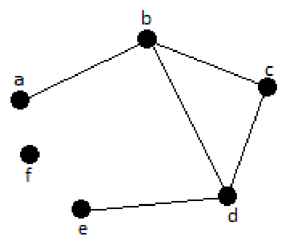
Note that we have defined an edge as a set of cardinality two: this means that a vertex cannot have an edge to itself, and there is no sense of direction in edges. Additionally, since edges are contained in a set, there can be at most one edge between any two vertices.

- If $u, v \in V(G)$, u is **adjacent to** v if $\{u, v\} \in E(G)$. (Note that by this definition a vertex is not adjacent to itself).
- The **degree** of a vertex is a count of the number of vertices it is adjacent to. Formally, for a vertex v, $\deg(v) = |\{u|\{v, u\} \in E(G)\}|$.
- The **empty graph** on n vertices has an empty edge set.
- The **complete graph** on n vertices K_n has all possible edges between vertices, that is, $E(G) = \{(u, v)|u, v \in E(G), u \neq v\}$. Every pair of vertices is adjacent.
- A **subgraph** G' of a graph G is a graph such that $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$. Since G' is a graph, each edge in $E(G')$ must be between two vertices in $V(G')$.

Task 4

Let G be a graph such that $V(G) = \{a, b, c, d, e, f\}$ and $E(G) = \{\{b, c\}, \{a, b\}, \{d, e\}, \{c, d\}, \{d, b\}\}$.

a. Draw G.



b. What is the degree of each vertex in G?

$\deg(a) = 1$, $\deg(b) = 3$, $\deg(c) = 2$, $\deg(d) = 3$, $\deg(e) = 1$, $\deg(f) = 0$

c. Say we take a vertex v uniformly at random from $V(G)$. Let X be a random variable that represents the degree of v. What is the probability distribution of X? That is, for each value in the range of X, what is the probability that X takes that value?

r	0	1	2	3
$\Pr(X = r)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{1}{6}$	$\frac{2}{6}$

d. What is $\mathbb{E}[X]$?

$\mathbb{E}[X] = 0 \cdot \frac{1}{6} + 1 \cdot \frac{2}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{2}{6} = \frac{2}{6} + \frac{2}{6} + 1 = \frac{5}{3}$

e. *Optional:* Can you develop a general expression for $\mathbb{E}[X]$ in terms of $|V|$ and $|E|$?

$\mathbb{E}[X] = \frac{2|E|}{|V|}$. The sum of all degrees is $2|E|$, and, by the linearity of expectation, we can do the calculation.

Checkoff — Call over a TA!